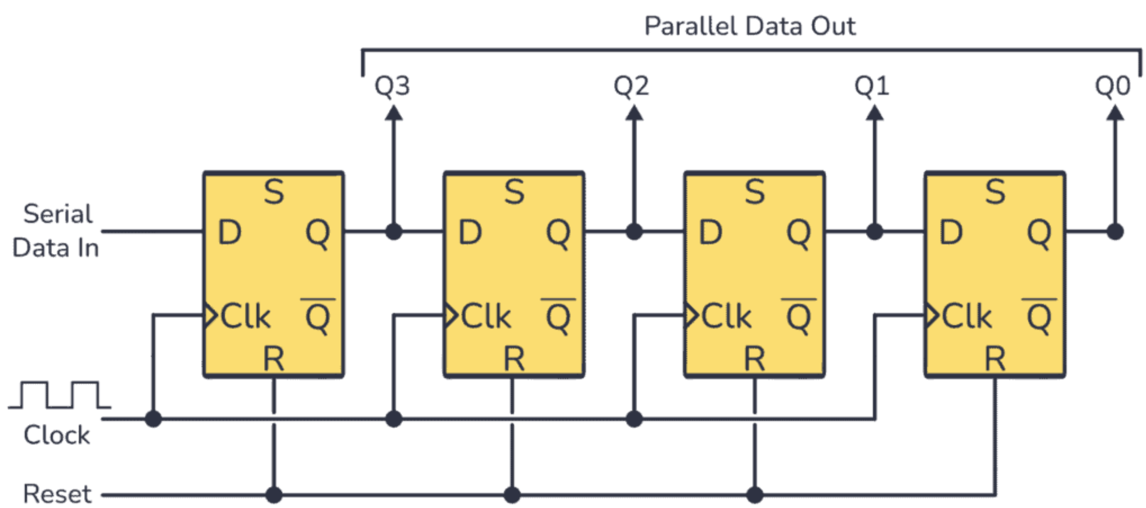
Day 31

Serial-In Parallel-Out (SIPO) Shift Register in Verilog

Circuit Diagram :



🔧 Verilog Code

4-bit Shift Register verilog code

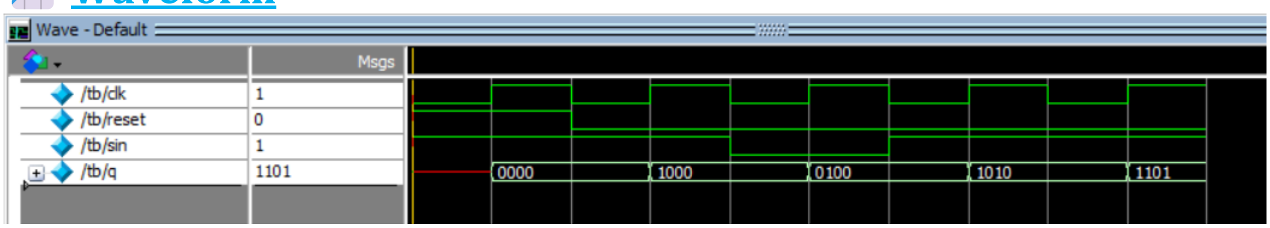
```
module SIPO(clk,reset,sin,q);
input clk,reset,sin;
output reg [3:0]q;
always @(posedge clk)
if(reset) q<=4'b0000;
else
begin
    q[3]<=sin;
    q[2]<=q[3];
    q[1]<=q[2];
    q[0]<=q[1];
end
endmodule
```

Testbench Code

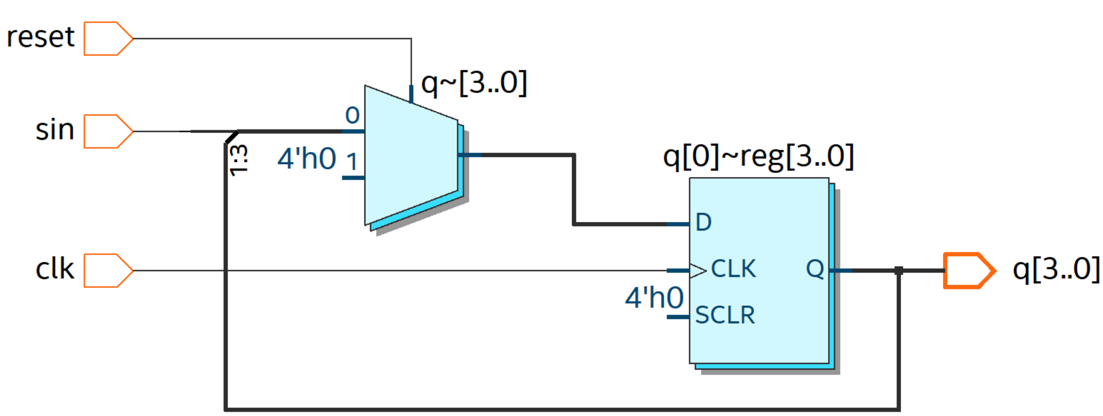
Tb code :

```
module tb;
reg clk,reset,sin;
wire [3:0]q;
SIPO dut(clk,reset,sin,q);
initial begin
clk=0;
forever #5 clk=~clk;
end
initial begin
reset=1;sin=1; #10
reset=0;sin=1; #10
sin=0; #10
sin=1; #10
sin=1; #10
$finish;
end
endmodule
```

Waveform



Schematic



🛠️ **EDA Tools Used**

- IntelQuartusPrime
- ModelSim